Paleomagnetism of the Ntonya Ring Structure, Malawi

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The natural remanent magnetization of specimens from eleven sites in syenitic and granitic rocks from the Ntonya ring structure is dominated by a large, randomly oriented IRM. Alternating-field demagnetization resulted in the isolation of a much smaller and stable component in specimens from seven sites. The mean direction of stable NRM was 311.3°E, +42.8° (downward), which corresponds on the geocentric dipole model to a paleomagnetic pole at 15.1°N, 27.8°W. At the other four sites either the stable component was too small relative to the IRM to be isolated from it by af demagnetization or it was not detected at all. The dispersion of stable NRM directions is considerably less than the dispersion associated with the present geomagnetic field. Because of its high stability against alternating-field and thermal demagnetization the systematically directed NRM is considered to be thermoremanence, acquired when the rocks cooled through the Curie point of the principle magnetic minerals (magnetite). Isotopic age determinations suggest that this stable NRM originated between 520 and about 600 m.y. ago, the latter being the most probable age. This study provides the first detailed result from late Precambrian or Lower Paleozoic rocks in Africa.

INTRODUCTION

There has been a gap in paleomagnetic results from Africa between about 1300 and 300 million years. Data from some thirty studies on Carboniferous and younger rocks and on older Precambrian formations are now adequate for the polar wander curve relative to the southern part of Africa to be traced with confidence from 300 m.y. to the present and tentatively from about 2700 to 1300 m.y. These data have been reviewed by *Gough et al.* [1964] and will be brought up to date by *McElhinny et al.* [1968]. The joining of the two portions of the polar wander curve has yet to be accomplished, because the only published results from the intervening period are from a single site in the Table Mountain series (Upper Silurian) near Cape Town [*Graham and Hales*, 1961], from a small collection of Cambrian volcanics in Morocco [*Helsley*, 1965], and from three sites in the Hook intrusives of Cambro-Ordovician age in western Zambia [*Brock*, 1967].

In attempts to bridge this gap extensive collections have been made of igneous rocks of presumed Lower Paleozoic age in Malawi, by the present author, and in Zambia, by A. Brock, and of red mudstones from the equivalent of the Table Mountain series in Natal (J. C. Briden and A. Brock, in *McElhinny et al.*, 1968). None of these yielded information on the paleomagnetic field except the Hook intrusives and the Ntonya ring structure, which is the subject of this paper.

The Ntonya ring structure lies about 5 km south of Zomba near the Blantyre road and is centered on 35°19′E, 15°27′S. It has been mapped by *Bloomfield* [1965] (Figure 1) and is considered on structural grounds to have been emplaced at the close of orogenic movements in the Mozambique belt, that is in the latest Precambrian or early Paleozoic, according to isotopic age studies. The complex is about 7 km in diameter and comprises a central coarse-grained intermediate rock of monzonitic composition, partly enveloped by a ring of foliated granite. The central unit, termed 'anatectic quartz-syenite' by Bloomfield, is not a simple igneous rock; it contains plagioclase and microperthite, quartz, biotite, amphibole, clinopyroxene, and hypersthene with auxiliary apatite, zircon, calcite, and iron ore, which is almost certainly magnetite. Complex intergrowths of quartz and feldspar, and sometimes quartz and biotite, are seen in some specimens.

Oriented samples were drilled at ten sites in the anatecitic quartz- syenite and at one site in the granite, using the drill described by

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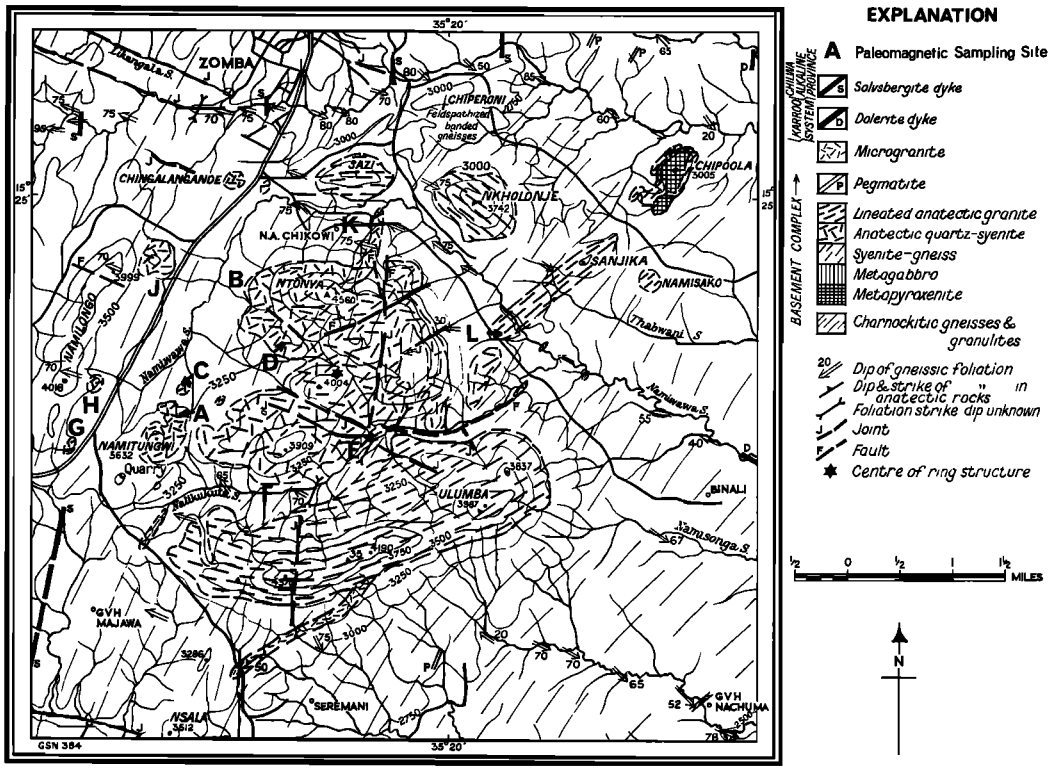


Fig. 1. Geology and paleomagnetic sampling sites of the Ntonya ring structure, reproduced, with additions, from *Bloomfield* [1965] by permission of the Commissioner of the Department of Geological Survey, Malawi Government, and the author.

McElhinny and Neal [1966] and an orienting system similar to that described by *Gough and Opdyke* [1963]. Checks showed that solar and magnetic bearings were equally reliable. From the cores, which were about 60 cm long, three cylindrical specimens were cut near the 30, 45, and 60 cm marks. At the first site (A), which was sampled in 1965, three cores were drilled; one specimen from each (at 30 cm) was subjected to alternating-field demagnetization and a further specimen from one core (at 60 cm) was treated by thermal demagnetization. At each of the other ten sites, which were sampled in 1966, two cores were drilled, and two specimens from each (30 and 60 cm) were treated in alternating fields and one of the specimens from 45 cm was thermally demagnetized. The af demagnetizing apparatus was that described by *McElhinny* [1966], and the thermal demagnetizer was that used by *Opdyke* [1964]. Direction and intensity of remanence were measured by the *Gough* [1964] spinner magnetometers.

Measurements of Total NRM

Untreated NRM directions were consistent within each separate core but differed greatly between cores. Site mean directions of total NRM at only five sites were statistically significant [*Fisher*, 1953; *Watson*, 1956; *Irving*, 1964] at the 95% confidence level, and even these probably have no physical meaning. When these five site means are combined, the resultant cannot be considered significant (Table 1). Intensities of total NRM were invariably greater in specimens from nearer the surface and decreased with depth. The great majority of the NRM is, therefore, ascribed to a surface effect, probably lightning, and is not of any paleomagnetic significance.

TABLE 1. Summary of Measurements of NRM before and after Magnetic Cleaning

Site	N	Total NRM R	Total NRM D, °E	Total NRM I, deg	Field Used for Magnetic Cleaning, mW m^{-2}	After Magnetic Cleaning R	D, °E	I, deg	k	α, deg	Pole ϕ	Pole λ	
A	3	2.66	110.6	+57.1	30	2.99	311.3	+45.4	161	9.7	13.2°W	26.5°N	
B	4	3.70	228.8	+53.7	45	3.95	311.4	+41.1	61	11.8	16.2°W	28.6°N	
C	4	1.65			60–80	3.80	308.4	+40.9	15	24.5	18.1°W	26.4°N	
D	4	2.71			30	3.99	311.6	+41.3	407	4.6	15.9°W	28.7°N	
E	4	3.96	245.6	−19.8	45	3.97	309.1	+44.6	93	9.6	15.1°W	25.4°N	
F	4	3.82	295.5	−30.3	30–45	3.88	(166.4	−73.1)	25	18.7	(28.0°E	15.0°N)[a]	
G	4	2.27			30–45	3.36	(270.2	+41.1)	5	47.8	(31.8°W	5.9°S)	
H	4	3.90	024.2	−02.3	45	3.98	314.5	+41.5	121	8.4	14.1°W	30.7°N	
J	4	1.78			45–60	3.95	312.6	+44.6	61	11.9	13.1°W	27.9°N	
K	4	2.45			30–45	3.19	(303.4	+37.8)	4	55.6	(22.8°W	23.5°N)	
L	4	2.25											
5 Sites		1.42			7 site mean directions	6.99	311.3	+42.8	1054	1.9	15.1°W	27.8°N	$d\psi = 1.4$ $d\chi = 2.3$
					7 site mean poles	6.99			$K = 1130$	$A = 1.8$	15.1°W	27.7°N	

Notes

N = number of specimens.
R = Resultant of N unit vectors; D, I are omitted where R is less than the 5% significance point of a randomness test [*Watson*, 1956].
D = Mean declination, degrees E of N.
I = Mean Inclination, degrees, downward positive.
k = Estimate by *Fisher* [1953] of precision of directions.
α = semiangle of cone of 95% confidence about the mean direction.
ϕ, λ are latitude and longitude of mean paleomagnetic north pole.
$d\psi$, $d\chi$ are semiaxes of oval of 95% confidence about the mean pole, respectively, along and perpendicular to the paleomeridian through the sampling site.
K = estimate by *Fisher* [1953] of precision of mean pole.
A = semiangle of cone of 95% confidence about the mean pole.
a = at site F the paleomagnetic south pole is given.

Alternating-Field Demagnetization

Sites A, B, C, D, E, H, J. Upon alternating-field demagnetization in progressively higher fields, NRM directions moved rapidly along great circles from their initial positions to a stable direction close to $D = 311.3°$, $I = +42.8°$ (Figure 2). This stable direction was reached after treatment in 30- to 80-mW m^{-2} (300 to 800 oe) peak field; further treatment up to 180 mW m^{-2} did not alter the direction appreciably. As direction changed through partial demagnetization, intensity dropped rapidly until the stable direction was reached when the remanence was of the order of 1% of its original magnitude; thereafter intensity decreased only slowly and exponentially with increased applied alternating field (Figure 3). These characteristics are interpreted in terms of a large isothermal remanence, which is easily removed by af demagnetization, underlain by a much smaller and very stable component. In an igneous rock, such a stable component that demagnetizes exponentially in alternating fields is usually considered to be thermoremanence-acquired when the rock has cooled through its Curie point. If the entire coercive force spectrum of thermoremanence is assumed exponential, the initial intensities of the stable component are calculated to be in the range 0.1 to 0.6 amp m^{-1}.

Site F. This site behaved similarly to the seven sites discussed above, except that the stable direction was completely different, at $D = 166.4°$, $I = -73.1°$ (Table 1). This is about 150° away from the group. The site is on a spur of a hill, and on aerial photographs it appears that it could have been a landslip. At any rate, there is sufficient doubt to force exclusion of this site from the analysis, and certainly it cannot be cited as evidence of field reversal. The possibility that it represents a stable NRM dating from a different time from that in the seven well-grouped sites is excluded because K/Ar analyses (this paper) do not show any age difference and there is no field evidence that site F is in a separate intrusion.

Site G. Directions moved along great circles toward the stable direction at the seven sites, but intensity dropped very rapidly and the stable component could not be isolated. If it is present, it is very weak. Directions varied randomly when an intensity of about 10^{-3} amp m^{-1}, or about one-half of 1% of total NRM, was reached.

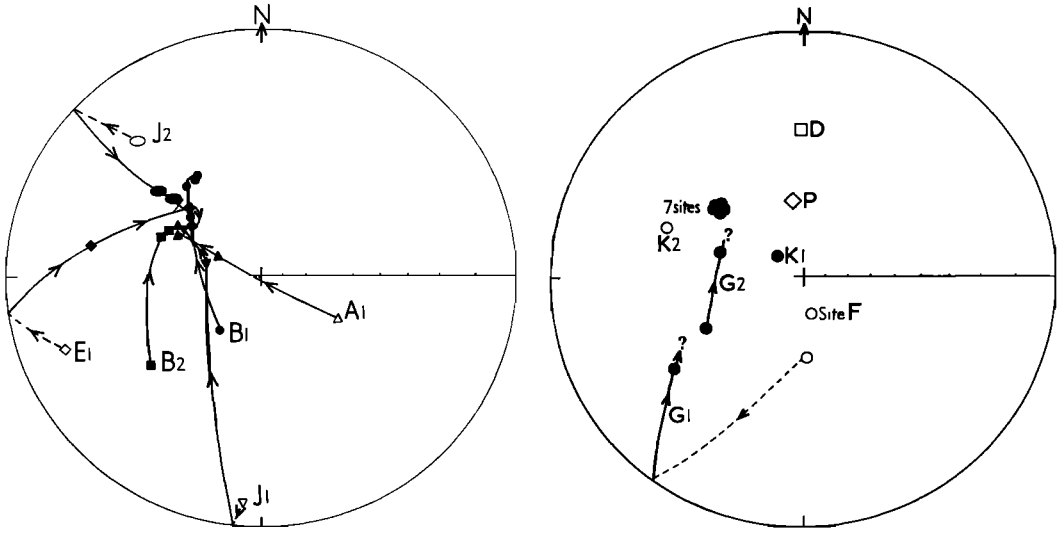


Fig. 2. Stereographic projections showing the change of direction of NRM with partial alternating-field demagnetization. Solid symbols denote downward inclinations, and open symbols denote upward inclinations. Demagnetization of one specimen of each of the cores labelled on the figure is illustrated. (*Left*) The mean direction of stable NRM at seven sites is marked with a cross. (*Right*) Directions at site K do not change appreciably through demagnetization; the stable end direction at site F is shown, and the present field and axial dipole field directions at the site are denoted by P and D.

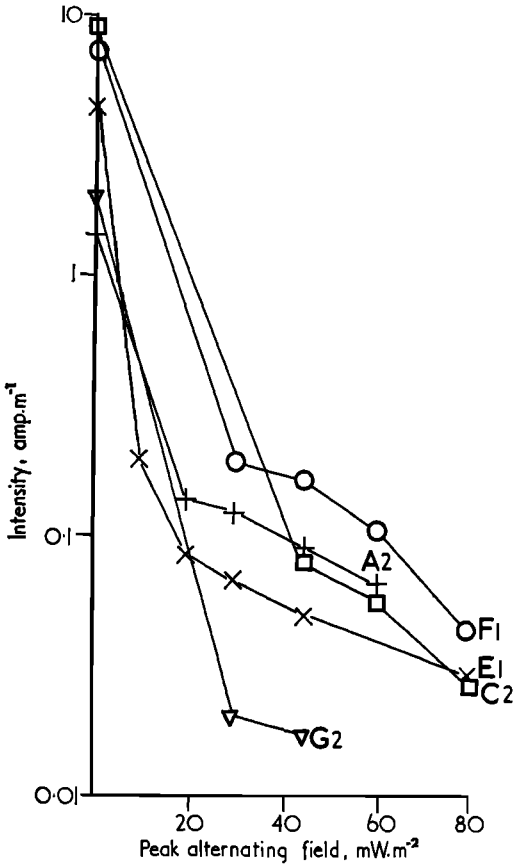


Fig. 3. Alternating field demagnetization curves.

Site K. Directions in two specimens from one core remained at around $D = 300°$, $I = +75°$ throughout demagnetization. In the other core there was some movement from initial directions around $D = 290°$, $I = -31°$ toward the stable group direction, but all systematically directed remanence was removed before the end direction could be reached. Even giving unit weight to specimens ($N = 4$), the mean of the last systematic directions was barely significant ($P = 0.05$) (Table 1), and probably N should be taken as 2, because the two cores behaved quite differently. Therefore, this site must be excluded from the analysis.

Site L. This site was in the foliated granite that surrounds the syenite. No systematic results were obtained: the NRM demagnetized steadily in alternating fields up to 60 mW m^{-2}, but there was no sign of the emergence of a significant site mean direction.

Thermal Demagnetization

Thermal demagnetization was not as successful in dissecting the components of remanence. At all the sites the demagnetization curves were approximately linear, which supports the conclusion already mentioned that the bulk of the remanence is isothermal. At sites A, B, C, D, E, H, and J, directions moved steadily toward the stable end direction indicated by the af treatment, but demagnetization close to the Curie point failed to remove all the secondary component: at 553°C the secondary and underlying primary components were of comparable magnitude (Figure 4). Extrapolation on the vector diagram of Figure 4 suggests that the magnitude of the stable component was initially of the order of 0.5 amp m^{-1}, which is consistent with the estimate from af demagnetization data. Site F again behaved in a similar manner to the group, and directions moved toward the discordant site mean direction which has been observed in the af study. Sites G and K, which had responded only sluggishly to alternating fields, were even less responsive to thermal treatment. No systematic results were obtained from the granite (site L). The usefulness of the thermal treatments was (1) that the bulk of the remanence was seen to have the characteristics of IRM, and (2) that the convergence of directions at the same seven sites was toward the same stable end direction, which had been derived from af measurements; this strongly supports the stability of the component in the direction $D = 311.3$, $I = +42.8°$.

Age of Stable NRM

Laboratory evidence indicates that the stable NRM is thermoremanence. To determine its age it is necessary to know the time when the intrusion cooled through the Curie point of the magnetic minerals (~570°C). The high strontium content of the Ntonya rocks (~800 ppm) compared with rubidium (~50 ppm) renders them unsuitable for Rb/Sr age determinations. K/Ar age determinations (Table 2) do not give a conclusive estimate of the age of intrusion. Two determinations on biotites give younger ages than hornblendes from the same samples. The range of hornblende ages is about 40 m.y., which is within experimental errors. The very low dispersion of stable NRM

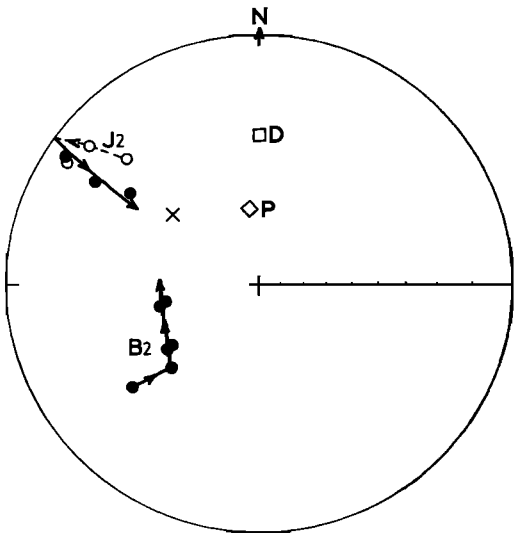


Fig. 4*a*.

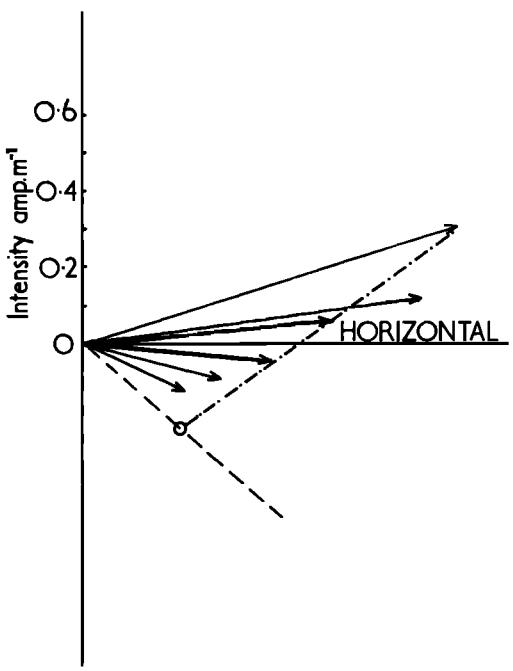


Fig. 4*b*.

Figure 4 shows the changes of direction during partial thermal demagnetization up to 550°C. (*a*) Stereographic projection for cores B2 and J2. The present and axial dipole field directions at Ntonya are denoted by P and D, and the stable mean direction for the Ntonya ring structure is denoted by a cross. (*b*) Vector diagram in the vertical plane bearing 310°E showing change in inclination and intensity of NRM through thermal demagnetization of a specimen from core J2.

directions in the Ntonya rocks suggests that the magnetism dates from a single age (within a few million years or less), when the temperatures were fairly uniform throughout the intrusion. This contrasts with the variable lower-temperature thermal history reflected in the spread of K/Ar ages and suggests that the preservation of magnetization and the argon retention date from different stages in the cooling history. It follows that the NRM is older than the K/Ar ages, which are to be regarded as minimum estimates. *Cahen and Snelling* [1966] have noted the association of circa 600 m.y. ages from several U/Pb and Pb-α analyses and from one Rb/Sr analysis in rocks from the Mozambique belt, which they consider to post-date the last major tectonic and metamorphic event in the belt. These rocks commonly yield K/Ar and Rb/Sr ages of circa 500 m.y. They regard the preponderance of K/Ar and Rb/Sr ages from the Mozambique belt in the range 550 to 450 m.y. (a characteristic evident in the Ntonya analyses) to be due to 'a long time lag between metamorphic crystallization and the time when the minerals formed during the metamorphism became closed systems with respect to argon and strontium diffusion.' The tectonic evidence suggests that the intrusion of the Ntonya ring structure and other similar complexes was associated with this last major phase of activity [*Bloomfield*, 1965]. The structure is, therefore, regarded as having a minimum age of 520 m.y., and the most probable age is about 600 m.y. The time of cooling through the Curie point is likely to have been closer to the time of intrusion than to the time recorded by the K/Ar ages. Unfortunately, the presence of a large IRM component prevents the detection of a viscous PTRM [*Chamalaun*, 1964; *Briden*, 1965], which would be expected if the rocks were maintained for tens of millions of years at elevated temperatures below the Curie point.

The paleomagnetism and age determinations rule out a metamorphic event at temperatures approaching the Curie point. Beyond this they do not help to discriminate between the view of *Clifford* [1967] that there was a major tectono-thermal event at about 500 m.y. and that of *Cahen and Snelling* [1966] that, if there was any event at that time (which they doubt), it was 'essentially mild' 'though possibly pervasive.'

TABLE 2. K/Ar Age Determinations from the Ntonya Ring Structure
Analyses by age determination unit, Institute of Geological Sciences.

Sample	Mineral	Wt % K	Ar_{rad}^{40} (scc/g)	% Ar_{rad}^{40}	Age, m.y.
NTH2	Biotite	5.79	1.2614×10^{-4}	96.8	479 ∓ 16
NTH2	Hornblende (+ pyroxene)	0.64	0.1628×10^{-4}	80.3	548 ∓ 85
NTF2	Hornblende (+ pyroxene)	0.99	0.2461×10^{-4}	75.7	537 ∓ 16
NTE2	Pyroxene (+ hornblende)	0.24	0.0575×10^{-4}	35.3	524 ∓ 30
NTB2	Biotite	6.87	1.5057×10^{-4}	96.7	474 ∓ 16
NTB2	Hornblende	1.31	0.3099×10^{-4}	94.7	515 ∓ 36

Discussion

Although the tight grouping of stable NRM directions suggests that the magnetism at the seven sites is essentially the same age, it would be surprising if they were synchronous within 10^5 years or so, because the rocks are medium- to coarse-grained and are likely to have cooled slowly and because the intrusion has been sampled from near its center to its edge. It seems likely, therefore, that the paleosecular variation at the time these rocks were magnetized is fully represented in the results. It is very much smaller than the secular variation associated with the present geomagnetic field—the paleolatitude of Ntonya is 25° and the dispersion $\Delta(= 81° \ k^{-1/2})$ is calculated as 2.5°, whereas, for the present field at latitude 25°, $\Delta \simeq 15°$ [*Creer*, 1962].

The intrusion is considered to have been emplaced at the close of major tectonic activity in the region; later faulting and igneous activity do not appear to have been accompanied by folding or regional tilting. Therefore, the mean paleomagnetic pole from the seven stable sites, which is located near the Canary Islands, is believed to be an accurate and representative pole relative to east-central Africa for about 600 m.y. ago (and, it will be argued later, to a much larger area as well).

This pole and its circle of 95% confidence lie within the circle of confidence around the pole deduced by *Brock* [1967] from the Hook intrusives, which are considered to be about 500 m.y. old on the basis of Rb/Sr determinations [*Snelling et al.*, 1964]. Therefore, there is no evidence of polar movement relative to southern Africa in the interval 600 to 500 m.y. These poles are, however, some 65° away from the pole deduced by *McElhinny and Opdyke* [1968] from some Carboniferous varvoids from central Africa.

On the basis of a reconstruction of Gondwanaland, which is compatible with all the

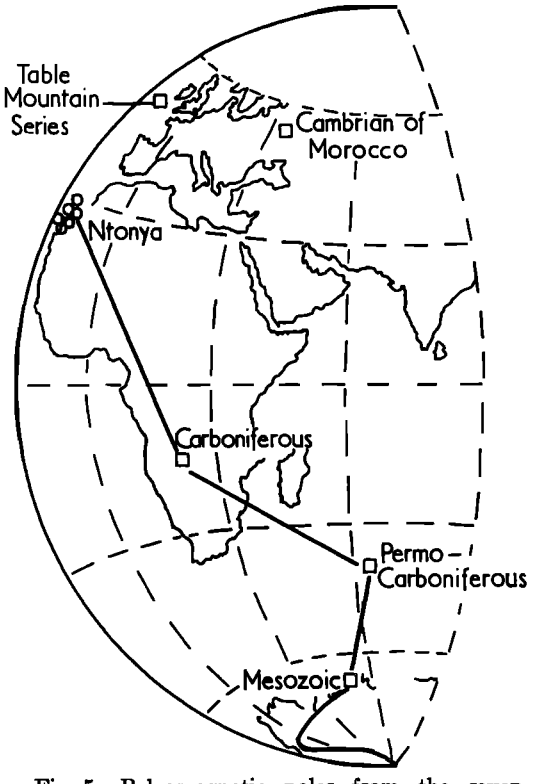


Fig. 5. Paleomagnetic poles from the seven stable sites of the Ntonya ring structure (denoted by circles) and previously published poles for Africa. A possible polar wander curve relative to Africa is drawn.

paleomagnetic data, *Briden* [1967] has suggested that the difference between the Carboniferous and early Paleozoic poles is due entirely to drift within the middle Devonian. On this view, pole positions similar to the pole from the Ntonya ring structure should be deduced from all African rocks in which the magnetism dates from 600+ to about 390 m.y. Indeed, the preliminary pole estimate from the late Silurian Table Mountain series at Cape Town [*Graham and Hales*, 1961] is only 24° away from the Ntonya pole, which may not be a significant difference.

The preliminary study on Cambrian lavas from Morocco [*Helsley*, 1965] yielded a pole at 53°N, 34°E, which contrasts with the two estimates from central Africa. There are three explanations of this difference: (1) The paleomagnetic pole made a large brief excursion during the middle Cambrian (there is no other evidence for this); (2) the Moroccan study has not yielded a typical estimate of the middle Cambrian paleomagnetic pole (which is quite likely as this was only a preliminary study); or (3) there have been vast relative movements between Morocco and central Africa since the Cambrian. If such relative movements have taken place, the boundary between the two crustal units must have been near Morocco, because, although Helsley's pole is consistent with the occurrence in Morocco, itself, of Cambrian archaeocyathinae, which are considered by *Irving* [1964] to be indicators of low to intermediate paleolatitude, it is not compatible with paleoclimatic evidence from the rest of Africa. Extensive Cambro-Ordovician glaciations in the Sahara [*Beuf et al.*, 1966] are in high paleolatitude and may be regarded as part of a polar ice cap, according to the paleomagnetic estimates from *central* Africa but *not* from Morocco. The pole deduced from the Ntonya ring structure is, therefore, regarded as a representative pole position for about 600 m.y. relative to the whole of Africa, with the possible exception of the northwestern margin. This northwestern margin, however, is regarded geologically to have been marginal to the African craton in the Lower Paleozoic, as it is now. Further, paleomagnetic study in Morocco is required to resolve these alternative conclusions.

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